Ahmad Sebaq

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Summary

Senior Computer Vision & AI engineer (7 years) specializing in generative models, vision-language models, and agentic LLM pipelines. Technical lead who has shipped research to production and led teams of up to 8.

Education

Nile University — Giza, Egypt
M.S. in Communication and Information Technology, GPA: 4.0/4.0 — 2021–2026

Minia University — Minya, Egypt
B.S. in Biomedical Engineering, Grade: Very Good with Honors (84.6%) — 2014–2019

Experience

CyShield — New Cairo, Egypt
Senior Computer Vision Engineer *(Full-time)* — January 2025 – Current

- **Biometric Border Control:** Built face- and fingerprint-based identity verification for border-control checkpoints — face detection, quality/liveness filtering, embedding extraction and fingerprint minutiae matching, with 1:1 verification and 1:N identification against watchlist galleries under strict latency and false-match constraints.
- **Stadium Video Surveillance:** Delivered a multi-camera stadium surveillance system — person detection, multi-object tracking and re-identification across overlapping camera views, plus action recognition for crowd-incident and prohibited-behaviour alerts on live video streams.
- **Generative Models:** Designed and trained diffusion- and GAN-based inpainting models for image restoration and content-aware editing, owning the full pipeline: dataset curation, mask generation, architecture selection, training at scale, and evaluation against perceptual (LPIPS, FID) and structural (SSIM, PSNR) metrics.
- **VLM Fine-tuning:** Fine-tuned vision-language models on domain-specific image-text pairs to improve grounding, captioning, and visual question-answering accuracy on in-house data. Built semantic search and retrieval pipelines over vector databases (embeddings indexing, similarity search, hybrid filters) to power RAG and visual retrieval workflows.
- **Technical Leadership:** Led a squad of 3 AI engineers — scoping work, code review, mentoring on generative-model best practices, and aligning research direction with product deadlines.

CNTXT — Remote — UAE / Egypt
AI/Computer Vision Consultant *(Part-time)* — January 2026 – Current

- **Robotics Data & Teleoperation:** Built an egocentric data-collection and QA platform processing 100+ hours of 3-camera teleoperation video per week into validated LeRobot datasets for imitation learning — 3D hand-pose reconstruction, automated video QA, and VLM-based action annotation. Auto-QA agreed with human reviewers on 90.8% of de-duplicated submissions, rejecting 0 of 333 good videos.
- **VLM Fine-tuning (SFT + RL):** Fine-tuned Qwen3-VL-8B with QLoRA (ms-swift, PEFT, NF4, single 24 GB L4) to segment and caption teleoperation video. On 497 held-out windows: segment-IoU F1 0.56 → 0.71, boundary-F1@0.5 s 0.18 → 0.39, caption token-F1 0.22 → 0.57. Added a GRPO stage with a metric-based reward; adapted to a new client from 15 episodes (boundary F1@1 s 0.12 → 0.58).
- **Agentic RAG (Arabic):** Co-developed an Arabic citation-grounded RAG service on LangGraph (intent router, multi-query RRF hybrid search, LLM evaluator–optimizer loop, streamed inline citations), deployed on AWS EKS. Owned retrieval quality: Qwen3 embedding and cross-encoder reranking lifted Hit@1 0.84 → 0.95 and nDCG@10 0.85 → 0.93; calibrated LLM judges against 200 human labels (weighted κ up to 0.75) and

redesigned grading to lift correct "no-answer" refusal 55% → 94%. Cut retrieval latency from up to 22.6 s to ∼1.3 s. *Stack:* FastAPI, Qdrant, GPT-4o, Postgres, Docker.

- **GPU Serving on AWS:** Built GPU inference services on AWS EKS and EC2 L4 — admission control, request de-duplication, result caching (23.3 s → 1 ms on repeats), SQS workers and S3-hosted LoRA adapters; sized 40K video-hours/month at 123–161 L4 workers.
- **Speech & Multi-modal CV:** Arabic speech pipeline (denoising, pyannote diarization, Whisper-large-v3 / OmniASR) benchmarked across 4 dialects (∼34% WER, ∼14% CER); Mask2Former instance segmentation (mask mAP 0.59 → 0.68, 200K annotations) and CLIP-based RGB/IR drone-frame matching.

Lumin Soft Giza, Egypt
AI Team Lead *(Full-time)* November 2023 – January 2025

- **Team Leadership:** Led a team of 8 AI developers across segmentation, detection, tracking, and OCR projects — running 1:1s, sprint planning, and technical mentorship.
- **AI Solution Development:** Delivered 3 production projects spanning segmentation, detection, tracking, clustering, re-identification (ReID), and OCR.
- **Sports Analytics (flagship):** Led a multi-model sports video pipeline — player/ball detection & tracking, team clustering, ReID, action recognition, possession tracking — output as annotated video plus metrics reports.
- **Research-to-Production:** Turned research prototypes into deadline-bound production pipelines.

Applied Innovation Center Giza, Egypt
R&D Engineer *(Full-time)* October 2022 – November 2023

- **Land Cover Detection:** Segmented land cover into 7 classes from Sentinel-2 and Planet satellite imagery using U-Net and Transformer networks, reaching 98% overall accuracy across Egypt's main governorates.
- **Geospatial Pipelines:** Built end-to-end satellite-imagery pipelines (preprocessing, tiling, model inference, post-processing), collaborating with GIS analysts on validation and deployment.

Center for Informatics Science, Nile University Giza, Egypt
Research Assistant *(Full-time)* April 2021 – April 2023

- **Disease Classification:** Classified rice crop diseases across 5 classes with ResNet and Transformer models, achieving 96% F1 on a 250K-image dataset.
- **Kaggle Competitions:** Participated in computer vision challenges, including UW-Madison Gastrointestinal Tract image segmentation.

Dawarha Cairo, Egypt
Computer Vision Engineer *(Part-time)* April 2021 – September 2023

- Built a real-time YOLOv3 + KMeans pipeline for 2-class recyclable-material detection (90 mAP), deployed on a Raspberry Pi at 11 FPS.

VLens Cairo, Egypt
Computer Vision Engineer *(Part-time)* February 2021 – November 2021

- Built national-ID OCR with U-Net text segmentation and KMeans preprocessing, deployed on AWS EC2.

Affectiva Massachusetts, USA
Mentor *(Contract, 3 months)* June 2020 – August 2020

- Mentored student teams through CV project scoping, model selection, and demo delivery.

TechnoIsland Cairo, Egypt
Computer Vision Engineer *(Part-time)* October 2019 – November 2021

- Built a CV pipeline for sperm-sample analysis — YOLOv3 detection, Yolact segmentation, DeepSort tracking across video frames.

Projects

BlacksmithAI *(Freelance, 2026)* blacksmithai.org
Natural-language to CAD agentic pipeline

- **Agentic pipeline:** Designed a multi-stage state machine (decompose → plan → assemble → execute → validate → repair) over a deterministic JSON IR with content-addressed caching, multi-modal image grounding, and human-in-the-loop review.
- **LLM tool-use & verification:** Built a Pydantic-validated tool catalog and pure verifier tools the LLM uses to self-check IR before execution; routed Claude Haiku and Gemini Flash via OpenRouter. *Stack:* FastAPI, Qdrant + BGE RAG, CadQuery, React + react-three-fiber, Docker.

TV Ad-Recognition Pipeline
Real-time broadcast ad detection via multi-modal fingerprinting

- Built a hybrid recognition system matching live TV broadcasts against an ad catalog using audio fingerprinting (Chromaprint), speaker embeddings (WavLM x-vectors), and scene-change keyframes (PySceneDetect + pHash + CLIP ViT-B/32 embeddings), fused over Elasticsearch kNN + BM25 search with NMS-suppressed airing events. Added a persistent-socket WebSocket endpoint for live per-channel recognition. *Stack:* two FastAPI services, Elasticsearch, PyTorch, FFmpeg; deployed as auto-start Windows services.

Sperm Analysis (CASA)
Computer-aided semen analysis desktop product

- **Analysis engine:** Built a CASA pipeline measuring sperm count, motility (per-cell kinematics via multi-object tracking), and morphology from microscopy and industrial-camera video — YOLO detection served through ONNX Runtime.
- **Product & deployment:** Shipped as an offline Windows desktop app — FastAPI backend (PyInstaller) wrapped in an Electron + React UI, industrial-camera capture (GenTL / Harvesters), encrypted model weights, HTTPS licensing service, and PDF clinical reports.

Research

Fatima Fellowship Stanford, USA
Research Intern *(Part-time)* September 2023 – August 2024

- Developed generative models to enhance low-resolution MRI images, focusing on model optimization and validation for clinical applications.

Linköping University Linköping, Sweden
Research Intern *(Part-time)* October 2022 – January 2023

- Developed GANs to manipulate Maxar aerial images by adding realistic seasonal effects, collaborating with geospatial analysts to ensure accuracy.

Publications

[1] T. A. Abdel Salam, A. M. Abdelsalam, N. R. Sholkamy, and **Sebaq, Ahmad**, "Assessment of artificial intelligence in detecting errors on panoramic radiographs", *Imaging Science in Dentistry*, vol. 56, e12, 2026. DOI: 10.5624/isd.20250242

[2] **Sebaq, Ahmad** and M. ElHelw, "Rsdiff: Remote sensing image generation from text using diffusion model", *Neural Computing and Applications*, vol. 36, no. 36, pp. 23 103–23 111, 2024. DOI: 10.1007/s00521-024-10363-3

[3] Y. A. Alnaggar, **Sebaq, Ahmad**, K. Amer, E. Naeem, and M. Elhelw, "Rice plant disease detection and diagnosis using deep convolutional neural networks and multispectral imaging", in *International Conference on Model and Data Engineering*, Springer, 2023, pp. 16–25. DOI: 10.1007/978-3-031-21595-7_2

[4] M. S. M. Bahgat and **Sebaq, A. M.**, "An Analytical Computational Algorithm for Solving a System of Multipantograph DDEs Using Laplace Variational Iteration Algorithm", in *Advances in Astronomy*, Jun. 2021, p. 16. DOI: 10.1155/2021/7741166

Awards

- Full Master Scholarship from Nile University 2021–2023
- 3rd Place, Global Monitoring for Environment and Security (GMES) & Africa Incubation Challenge 2021
- 3rd Place in Affectiva Makeathon with Inside Out project 2020
- Nokia Africathon Finalist with ALS project 2019

Skills

- **Programming:** Python, C++, JavaScript / TypeScript, React
- **Generative AI & LLMs:** LLM orchestration (LangGraph, LangChain), multi-agent pipelines, tool-use / function-calling, agentic RAG, multi-modal VLMs, local inference (llama-cpp-python), vector & hybrid search (Qdrant, Elasticsearch; dense + BM25, RRF fusion, cross-encoder reranking, ColBERT late interaction), LLM-as-judge evaluation, prompt engineering
- **Computer Vision:** OpenCV; segmentation (U-Net, Mask R-CNN, Swin), detection & tracking (YOLO, Yolact, DeepSort), OCR, re-identification, 3D hand-pose / SLAM, CLIP cross-modal matching
- **ML / Deep Learning:** PyTorch, ONNX Runtime, HuggingFace, scikit-learn; CNNs, Transformers, diffusion models, GANs; ASR (Whisper), speaker diarization (Pyannote), denoising (DeepFilterNet3), audio fingerprinting
- **MLOps & Deployment:** FastAPI, Docker, Jenkins CI, AWS (EC2), Git; Postgres + SQLAlchemy/Alembic, JWT auth, structured logging & Sentry; desktop packaging (Electron, PyInstaller); edge deployment (Jetson Nano & Orin, Raspberry Pi)
- **Robotics & Geospatial:** LeRobot imitation-learning data pipelines, industrial-camera capture; QGIS, Sentinel-2 / Planet / Maxar satellite imagery